

McMaster Micromouse Competition Rules 2026

General Overview

Welcome to the 2026 McMaster Micromouse Competition! The objective is to build an autonomous robot (a “mouse”) that can independently navigate a complex maze to find the center destination in the shortest time possible.

- Teams may consist of one to five members.
- These rules are adapted from the official IEEE Micromouse specifications with minor local adjustments.
- **Please note: Terms highlighted in red are provisional. Final values and parameters may be adjusted based on event logistics and registration numbers.**

Robot Specifications

- **Originality:** The robot must be designed and built from scratch by the competing team members.
- **Autonomy:** The robot must be entirely self-contained and autonomous. Remote controls, tethers, or external communication devices are strictly prohibited.
- **Dimensions:** The robot’s physical footprint must remain within a 25 cm x 25 cm area at all times. There are no height restrictions.
- **Structural Integrity:** The robot cannot separate, deploy payloads, or leave any parts of itself behind while navigating. The entire robot must arrive at the destination square as one unit to successfully complete the maze.
- **Maze Preservation:** The robot must not jump over, fly above, climb, scratch, cut, burn, mark, damage, or otherwise destroy the walls or floor of the maze. Traction systems that cause damage to the maze surface are forbidden.

Maze Specifications

- **Dimensions:** The maze is laid out on a 16 x 16 unit grid. Each individual unit square measures 18 cm x 18 cm. The entire maze is securely enclosed by a perimeter wall.
- **Colors and Materials:** The walls feature white sides and red tops. The maze floor is painted black to aid in sensor contrast.
- **Start and Finish:** The starting position is located at one of the four corners of the maze. The final destination is located in the center of the maze layout.

Competition & Run Rules

- **Time Allocation:** Each team is allotted a total of 10 minutes inside the maze. During this window, the robot may make multiple autonomous runs to explore, map the maze, and execute fast speed-runs.
- **Run Timing:** The official run time is tracked using a start line (exiting the starting square into the next unit square) and a finishing line (the entrance threshold to the destination square). The team's best (shortest) run time will be recorded as their final score.
- **Blind Navigation:** Information regarding the specific maze layout **CANNOT** be downloaded, programmed into, or given to the robot beforehand. It must map the maze dynamically.
- **Permitted Adjustments:** Once the maze is revealed, teams may physically touch their robot between runs to make the following adjustments:
 1. Changing switch/button settings (e.g., selecting different search algorithms)
 2. Replacing or recharging batteries
 3. Adjusting physical sensor alignments
 4. Changing speed profile settings
 5. Performing minor mechanical repairs
- **Interventions:** Teams may safely abort and restart their runs at any time. If the robot becomes immobilized, stuck in a corner, or caught on a wall, the team may manually intervene to rescue the robot. However, they may not manually point or alter the robot's intended direction of movement to assist its navigation.

Scoring & Awards

- **Top Placements:** First, second, and third place will be awarded to the teams achieving the shortest official run times. *Note: To qualify for these top placements, the robot must complete its winning run continuously, without the team touching or intervening with the robot during that specific run.*
- **Honourable Mentions:** We will also recognize teams for creativity and engineering with honourable mentions in the following categories:
 - [TBD]

Good luck and have fun everyone!